MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEECOA01 - INVESTMENT ENVIRONMENT AND PORTFOLIO MANAGEMENT STRATEGIES

QP CODE : 25901221

Part A: Short Answer Questions

Answer any 10 questions. Each carry 2 marks (2×10 = 20)

1 Options are derivative contracts that give the buyer the **right but not the obligation** to buy (Call option) or sell (Put option) an underlying asset at a predetermined strike price on or before a specific date. Their basic functions are **hedging** (protecting against price risk), **speculation** (betting on price movements with limited downside), and **income generation** (writing options for premium). Unlike forwards/futures, there is no obligation to exercise.

2 Financial leverage (debt-equity ratio), Operating leverage (fixed vs variable costs), Business/Industry risk (cyclical vs non-cyclical), Company size and market position, Economic conditions and interest-rate sensitivity.

3 Beta (β) is the measure of **systematic (market) risk** of a security or portfolio relative to the market portfolio. In the CAPM equation: **Expected Return = Rf + β (Rm – Rf)** $\beta = 1 \rightarrow$ moves with market; $\beta > 1 \rightarrow$ more volatile; $\beta < 1 \rightarrow$ less volatile.

4 **Call** and **Put** options, **American** (exercisable any time before expiry) and **European** (exercisable only on expiry)

5 A swap involves **two counterparties** who agree to exchange cash flows. In an interest-rate swap, one party (fixed payer) pays a fixed rate and receives a floating rate; the other (floating payer) does the opposite. No principal is exchanged (notional principal only). Parties are typically a bank/financial institution and a corporate client.

6 **Short-term** (days to a few months). Speculators aim to profit from short-term price fluctuations rather than long-term fundamentals.

7 Hedging (risk reduction), Speculation (leveraged bets), Arbitrage (exploiting mispricing), Portfolio insurance / dynamic asset allocation, Enhancing returns / income generation

8 The slope of the **Capital Market Line (CML)** is **(E(Rm) – Rf) / σm** (market Sharpe ratio). It represents the **additional expected return per unit of total risk** (standard deviation) for efficient portfolios combining the risk-free asset and market portfolio. Steeper slope = better risk-return trade-off.

9 As per SEBI (Portfolio Managers) Regulations, a **portfolio manager** is a **body corporate** registered with SEBI who, pursuant to a contract or arrangement with a client, **advises or directs or undertakes** on behalf of the client (whether discretionary or non-discretionary) the **management or administration of a portfolio of securities or the funds** of the client.

10 Variance is a statistical measure of the **dispersion of returns** around the mean (expected) return. It is the square of standard deviation and quantifies **total risk** (systematic + unsystematic) of an investment. Formula: $\sigma^2 = \Sigma (Ri - \bar{R})^2 / n$.

11 Personal goals (retirement, education, home purchase, etc.) determine **time horizon, risk tolerance, liquidity needs, return requirements, and tax status**. They guide asset allocation (e.g., equity for growth, debt for safety) to ensure investments match life-stage objectives.

12 Spreading investments across uncorrelated or low-correlated assets/sectors to **reduce unsystematic (specific) risk** without sacrificing expected return. Core of Modern Portfolio Theory – "Do not put all eggs in one basket."

13 Also known as **Dow Theory**, it is a technical analysis framework stating that the market moves in **three trends** (primary, secondary, minor). A primary trend is confirmed only when both the **Dow Jones Industrial Average** and **Transportation Average** move in the same direction. The market discounts all known information.

14 The process of **choosing the optimal combination of securities** that maximises return for a given risk level (or minimises risk for a given return) based on investor objectives, usually using Markowitz's efficient frontier or other optimisation models.

15 Investors are **rational, risk-averse**, hold fully diversified portfolios, and care only about **expected return and systematic risk (beta)**. They have homogeneous expectations and can borrow/lend at the risk-free rate.

Part B: Short Essay Questions

Answer any 6 questions. Each carry 5 marks (5×6 = 30)

16 Determine investment objectives & constraints (risk, return, time horizon), Security analysis (fundamental/technical/economy-industry-company), Portfolio construction (asset allocation, diversification), Portfolio revision/rebalancing, Portfolio performance evaluation.

17 Planning → Execution → Monitoring & Rebalancing → Evaluation.

18

Aspect	**Sharpe's Ratio**	**Treynor's Ratio**
Risk measure	Total risk (σp)	Systematic risk (βp)
Formula	(Rp – Rf) / σp	(Rp – Rf) / βp
Suitable for	Undiversified / total portfolio	Well-diversified portfolios
Focus	Excess return per unit total risk	Excess return per unit market risk

19 Standardised contracts on exchange; margin (initial + maintenance); daily mark-to-market; physical/cash settlement; no counter-party risk due to clearing corporation.

20 Calculate portfolio return, Risk-adjust performance (Sharpe, Treynor, Jensen), Performance attribution (sector/stock selection), Benchmark comparison, Review & corrective action.

21 Underwriting, M&A advisory, trading, research. NASDAQ = OTC electronic market. Depository settlement = DTCC/DTC (US) / NSDL/CDSL (India) for demat & T+1 settlement.

22 Questionnaire → Assess risk capacity & tolerance → Map to investor type (conservative/aggressive) → Recommend asset allocation. Benefits & limitations.

23 **$\alpha = Rp - [Rf + \beta (Rm - Rf)]$** Positive α = outperformance.

24 Maintains target asset allocation; controls risk drift; locks in gains; tax harvesting; aligns with changing goals.

25 Risk = uncertainty of actual vs expected return. **Systematic** (market, interest rate, inflation, purchasing power). **Unsystematic** (business, financial, liquidity).

Part C: Essay Questions

Answer any 2 questions. Each carry 10 marks (10×2 = 20)

26 Safety, Liquidity, Return, Tax benefits, Convenience, Marketability. **Types of investors**: Individuals (conservative/aggressive), Institutional (mutual funds, insurance, pension), Foreign (FIIs), Corporate.

27 Markowitz model → Efficient frontier → Risk reduction through low/negative correlation. Limitations (does not eliminate systematic risk). Numerical illustration (2-asset portfolio variance formula) recommended.

28 Prices reflect all available information → no abnormal returns possible. **Weak form**: Past price/volume info already reflected. **Semi-strong form**: All public information reflected. **Strong form**: All information (public + private) reflected. Implications for technical/fundamental analysis.

29 **Portfolio Management**: Focused on securities selection, asset allocation, risk-return optimisation. **Wealth Management**: Holistic – includes portfolio + tax planning, estate planning, insurance, retirement, philanthropy. **Differences** (scope, clients, services, regulation).